數統導論作業六

111020003 林群佑

4.2.8

By CLT we have :

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Then, by Z table we have :

$$0.954 \approx P\left(-2 < Z < 2\right)$$

Therefore, we can compute that:

$$\frac{1}{2} \approx 2\frac{\sqrt{10}}{\sqrt{n}} \implies n \approx 160$$

4.2.10

(1) Find the constant c

Let $Y = \bar{X} - X_{n+1}$. Since $\bar{X}$ and X_{n+1} are independent:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right), \quad X_{n+1} \sim N(\mu, \sigma^2)$$

The expectation and variance of Y are:

$$E[Y] = \mu - \mu = 0$$

$$Var(Y) = Var(\bar{X}) + Var(X_{n+1}) = \frac{\sigma^2}{n} + \sigma^2 = \sigma^2\left(\frac{n+1}{n}\right)$$

Standardizing Y gives a standard normal variable Z:

$$Z = \frac{(\bar{X} - X_{n+1}) - 0}{\sqrt{\sigma^2\left(\frac{n+1}{n}\right)}} = \frac{\bar{X} - X_{n+1}}{\sigma\sqrt{\frac{n+1}{n}}} \sim N(0,1)$$

We define the T-statistic with $n-1$ degrees of freedom as $T = \frac{Z}{\sqrt{S^2/\sigma^2}}$:

$$T = \frac{\frac{\bar{X} - X_{n+1}}{\sigma\sqrt{\frac{n+1}{n}}}}{S/\sigma} = \frac{\bar{X} - X_{n+1}}{S\sqrt{\frac{n+1}{n}}} \sim t(n-1)$$

The problem requires the form $c(\bar{X} - X_{n+1})/S$. Comparing the coefficients:

$$c = \frac{1}{\sqrt{\frac{n+1}{n}}} = \sqrt{\frac{n}{n+1}}$$

(2) find k for the 80 precentage prediction interval

We want to find k such that:

$$P(\bar{X} - kS < X_{n+1} < \bar{X} + kS) = 0.80 \implies P\left(-k < \frac{\bar{X} - X_{n+1}}{S} < k\right) = 0.80$$

From part (1) we know that T follows a t-distribution with $n-1$ degrees of freedom:

$$T = \frac{\bar{X} - X_{n+1}}{S\sqrt{\frac{n+1}{n}}} \sim t(n-1)$$

Since:

$$0.8 = P\left(-t_{0.1,7} < \sqrt{\frac{8}{9}}\frac{\bar{X} - X_9}{S} < t_{0.1,7}\right)$$

By T-table we have :

$$t_{0.10,7} \approx 1.415$$

Therefore :

$$k = t_{0.10,7} \times \sqrt{\frac{n+1}{n}} \approx 1.415 \times \sqrt{\frac{9}{8}} \approx 1.5$$

4.2.18

(a)

By question, we have:

$$P\left(a < \frac{(n-1)S^2}{\sigma^2} < b\right) = 0.95 \text{ and } P\left(\frac{(n-1)S^2}{\sigma^2} < b\right) = 0.975 \implies P\left(\frac{(n-1)S^2}{\sigma^2} \le a\right) = 0.025$$

Then chi-square is define on $[0, \infty)$, so $a > 0$. Because we have $a > 0, b > 0$:

$$P\left(a < \frac{(n-1)S^2}{\sigma^2} < b\right) = P\left(\frac{1}{b} < \frac{\sigma^2}{(n-1)S^2} < \frac{1}{a}\right) = P\left(\frac{(n-1)S^2}{b} < \sigma^2 < \frac{(n-1)S^2}{a}\right) = 0.95$$

(b)

Because $(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$:

$$a = \chi^2_{8,0.025} \approx 2.180, \; b = \chi^2_{8,0.975} \approx 17.534$$

From part (a) we know that :

$$P\left(\frac{(n-1)S^2}{b} < \sigma^2 < \frac{(n-1)S^2}{a}\right) = 0.95$$

Therefore, we have confidence interval for σ^2 :

$$\left(\frac{(n-1)S^2}{b}, \frac{(n-1)S^2}{a}\right) \approx \left(\frac{8 \times 7.93}{17.534}, \frac{8 \times 7.93}{2.180}\right) \approx (3.62, 29.1)$$

(c)

If the true population mean μ is known, we do not need to estimate it using the sample mean $\bar{X}$.

1. Change in Pivotal Quantity: Instead of using the sample variance $S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$, we use the quantity $\sum(X_i - \mu)^2$. The new pivotal quantity is:

$$\frac{\sum_{i=1}^{n}(X_i - \mu)^2}{\sigma^2}$$

2. Change in Distribution: Because we are using the true parameter μ rather than an estimate $\bar{X}$, we do not lose a degree of freedom. This quantity follows a Chi-square distribution with n degrees of freedom.

$$\frac{\sum(X_i-\mu)^2}{\sigma^2} \sim \chi_n^2$$

3. Modified Confidence Interval Formula: The confidence interval becomes:

$$\left(\frac{\sum_{i=1}^n (X_i-\mu)^2}{b}, \frac{\sum_{i=1}^n (X_i-\mu)^2}{a}\right)$$

where a and b are the critical values from the χ_n^2 distribution

4.2.22

We use CLT to estimate confidence interval for $p_1 - p_2$:

$$\hat{p_1} = \frac{y_1}{n_1} = \frac{50}{100} = 0.5,\ \hat{p_2} = \frac{y_2}{n_2} = \frac{40}{100} = 0.4 \implies \hat{p_1} - \hat{p_2} = 0.1$$

$$S^2 = Var\left(\hat{p_1} - \hat{p_2}\right) = \frac{n_1\hat{p_1}(1-\hat{p_1})}{n_1^2} + \frac{n_2\hat{p_2}(1-\hat{p_2})}{n_2^2} = \frac{0.5\times 0.5}{100} + \frac{0.4\times 0.6}{100} = 0.0049$$

$$\implies S = \sqrt{0.0049} = 0.07$$

Therefore, approximate 90 percentage confidence interval for $p_1 - p_2$ is :

$$(\hat{p_1} - \hat{p_2} - Z_{0.05}\times S, \hat{p_1} - \hat{p_2} + Z_{0.05}\times S) \approx (0.1 - 1.645\times 0.07, 0.1 + 1.645\times 0.07) \approx (-0.015, 0.215)$$

The approximate 90 percentage confidence interval for $p_1 - p_2$ is:

$$(-0.015, 0.215)$$

4.2.27

(a)

By definition we know that:

1. Samples X from $N(\mu_1, \sigma_1^2)$ its standardized variance follow a chi-squared distribution:

$$V = \frac{(n-1)S_1^2}{\sigma_1^2} \sim \chi^2(n-1)$$

2. Samples Y from $N(\mu_2, \sigma_2^2)$ its standardized variance follow a chi-squared distribution:

$$U = \frac{(m-1)S_2^2}{\sigma_2^2} \sim \chi^2(m-1)$$

3. Since X and Y are independent, U and V are also independent.

By definition of problem F is :

$$F = \frac{\frac{(m-1)S_2^2}{\sigma_2^2}/(m-1)}{\frac{(n-1)S_1^2}{\sigma_1^2}/(n-1)} = \frac{S_2^2/\sigma_2^2}{S_1^2/\sigma_1^2}$$

Therefore, we know that F fits the definition of F-distribution with numerator degrees of freedom $m-1$ and denominator degrees of freedom $n-1$. That is :

$$F \sim F(m-1, n-1)$$

(b)

By question, we have:

$$P\left(a < F < b\right) = 0.95 \text{ and } P\left(F < b\right) = 0.975 \implies P\left(F \leq a\right) = 0.025$$

Therefore:

$$a = F_{0.025}(m-1, n-1),\, b = F_{0.975}(m-1, n-1)$$

(c)

$$0.95 = P\left(a < F < b\right) = P\left(a < \frac{S_2^2\sigma_1^2}{S_1^2\sigma_2^2} < b\right)$$

Then $S_1^2 > 0, S_2^2 > 0 \implies S_2^2/S_1^2 > 0$ we have:

$$P\left(a < \frac{S_2^2\sigma_1^2}{S_1^2\sigma_2^2} < b\right) = P\left(a < \frac{S_2^2}{S_1^2} \cdot \frac{\sigma_1^2}{\sigma_2^2} < b\right) = P\left(a\frac{S_1^2}{S_2^2} < \frac{\sigma_1^2}{\sigma_2^2} < b\frac{S_1^2}{S_2^2}\right) = 0.95$$

4.4.6

(a)

$$P\left(min(X_1, X_2, X_3) > m\right) = P\left(X_1 > m, X_2 > m, X_3 > m\right)$$

Then, since X_1, X_2, X_3 are independent and by definition of median $P(X_i > m) = 1/2$:

$$P\left(min(X_1, X_2, X_3) > m\right) = P\left(X_1 > m\right)P\left(X_2 > m\right)P\left(X_3 > m\right) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$$

(b)

First we compute cdf of X:

$$F_X(x) = \int_0^x 2xdx = \left[x^2\right]_0^x = x^2,\, x \in (0,1)$$

Then we have :

$$\mathbb{E}[Y_2] = \int_0^1 x\frac{3!}{1!\,1!}(x^2)^1 2x(1-x^2)^1 dx = \int_0^1 12x^4 - 12x^6 dx = \left[\frac{12}{5}x^5 - \frac{12}{7}x^7\right]_0^1 = \frac{24}{35}$$

$$\mathbb{E}[Y_2^2] = \int_0^1 x^2\frac{3!}{1!\,1!}(x^2)^1 2x(1-x^2)^1 dx = \int_0^1 12x^5 - 12x^7 dx = \left[2x^6 - \frac{3}{2}x^8\right]_0^1 = \frac{1}{2}$$

$$Var\left(Y_2\right) = \mathbb{E}[Y_2^2] - \mathbb{E}[Y_2]^2 = \frac{1}{2} - \left(\frac{24}{35}\right)^2 = \frac{73}{2450}$$

$$\mathbb{E}[Y_3] = \int_0^1 x\frac{3!}{2!\,0!}(x^2)^2 2x(1-x^2)^0 dx = \int_0^1 6x^6 dx = \left[\frac{6}{7}x^7\right]_0^1 = \frac{6}{7}$$

$$\mathbb{E}[Y_3^2] = \int_0^1 x^2\frac{3!}{2!\,0!}(x^2)^2 2x(1-x^2)^0 dx = \int_0^1 6x^7 dx = \left[\frac{3}{4}x^8\right]_0^1 = \frac{3}{4}$$

$$Var\left(Y_3\right) = \mathbb{E}[Y_3^2] - \mathbb{E}[Y_3]^2 = \frac{3}{4} - \left(\frac{6}{7}\right)^2 = \frac{3}{196}$$

$$\begin{aligned}
\mathbb{E}[Y_2Y_3] &= \int_0^1 \int_0^{y_3} (y_2y_3) \cdot \frac{3!}{1!0!0!}(y_2^2)^1(y_3^2-y_2^2)^0(1-y_3^2)^0(2y_2)(2y_3)\, dy_2\, dy_3 \\
&= \int_0^1 \int_0^{y_3} (y_2y_3) \cdot 24y_2^3y_3\, dy_2\, dy_3 \\
&= 24\int_0^1 y_3^2 \left[\int_0^{y_3} y_2^4\, dy_2\right] dy_3 \\
&= 24\int_0^1 y_3^2 \left[\frac{y_3^5}{5}\right]_0^{y_3} dy_3 \\
&= \frac{24}{5}\int_0^1 y_3^7\, dy_3 \\
&= \frac{24}{5}\cdot\frac{1}{8} = \frac{3}{5}
\end{aligned}$$

Therefore, correlation between Y_2 and Y_3:

$$\rho_{Y_2,Y_3} = \frac{\mathbb{E}[Y_2Y_3]-\mathbb{E}[Y_2]\mathbb{E}[Y_3]}{\sigma_{Y_2}\sigma_{Y_3}} = \frac{3/5 - 24/35\cdot 6/7}{\sqrt{73/2450\cdot 3/196}} = \frac{2\sqrt{438}}{73} \approx 0.573$$

4.4.12

First, we have $f(x) = 2x$, $x \in (0,1)$, so the joint pdf of Y is :

$$f_{Y_1,Y_2,Y_3}(y_1,y_2,y_3) = 3!(2y_1)(2y_2)(2y_3) = 48y_1y_2y_3,\ \text{ where } 0 < y_1 < y_2 < y_3 < 1$$

Define Z_1, Z_2, Z_3 :

$$Z_1 = \frac{Y_1}{Y_2},\ Z_2 = \frac{Y_2}{Y_3},\ Z_3 = Y_3 \implies Y_1 = Z_1Z_2Z_3, Y_2 = Z_2Z_3, Y_3 = Z_3$$

Compute Jacobian matrix :

$$J = \det\begin{pmatrix} \frac{\partial y_1}{\partial z_1} & \frac{\partial y_1}{\partial z_2} & \frac{\partial y_1}{\partial z_3} \\ \frac{\partial y_2}{\partial z_1} & \frac{\partial y_2}{\partial z_2} & \frac{\partial y_2}{\partial z_3} \\ \frac{\partial y_3}{\partial z_1} & \frac{\partial y_3}{\partial z_2} & \frac{\partial y_3}{\partial z_3} \end{pmatrix} = \det\begin{pmatrix} z_2z_3 & z_1z_3 & z_1z_2 \\ 0 & z_3 & z_2 \\ 0 & 0 & 1 \end{pmatrix} = z_2z_3^2$$

Then we have :

$$f_{Z_1,Z_2,Z_3}(z_1,z_2,z_3) = f_{Y_1,Y_2,Y_3}(z_1z_2z_3,\ z_2z_3,\ z_3)\cdot|J| = 48z_1z_2^3z_3^5$$

Find domain of Z_1, Z_2, Z_3 :

$$\begin{aligned}
y_1 < y_2,\ \text{so } \frac{Y_1}{Y_2} < 1 &\implies 0 < Z_1 < 1 \\
y_2 < y_3,\ \text{so } \frac{Y_2}{Y_3} < 1 &\implies 0 < Z_2 < 1 \\
y_3 < 1 &\implies 0 < Z_3 < 1
\end{aligned}$$

Since domain of Z_1, Z_2, Z_3 are independent and it is obvious that joint pdf can be factorize :

$$f_{Z_1,Z_2,Z_3}(z_1,z_2,z_3) \propto (z_1)\cdot(z_2^3)\cdot(z_3^5)$$

By Factorization Theorem, Z_1, Z_2, Z_3 are mutually independent.

4.4.15

(a)

The pdf of $N(0,\sigma^2)$:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-x^2/2\sigma^2}$$

Then compute $\mathbb{E}[Y_1]$:

$$
\begin{aligned}
\mathbb{E}[Y_1] &= \int_{-\infty}^{\infty}\int_{-\infty}^{y_2} y_1 \frac{2!}{2\pi\sigma^2} e^{-(y_1^2+y_2^2)/2\sigma^2}\, dy_1\, dy_2 \\
&= \int_{-\infty}^{\infty} \frac{1}{\pi\sigma^2} e^{-y_2^2/2\sigma^2} \left(\int_{-\infty}^{y_2} y_1 e^{-y_1^2/2\sigma^2} dy_1\right) dy_2 \\
&\text{Let } u = \frac{y_1^2}{2\sigma^2} \implies du = \frac{y_1}{\sigma^2} dy_1 \\
&= \int_{-\infty}^{\infty} \frac{1}{\pi\sigma^2} e^{-y_2^2/2\sigma^2} \left(\int_{\infty}^{y_2^2/2\sigma^2} y_1 e^{-u} \frac{\sigma^2}{y_1} du\right) dy_2 \\
&= \int_{-\infty}^{\infty} \frac{1}{\pi} e^{-y_2^2/2\sigma^2} \left[-e^{-u}\right]_{\infty}^{y_2^2/2\sigma^2} dy_2 \\
&= \int_{-\infty}^{\infty} -\frac{1}{\pi} e^{-y_2^2/\sigma^2} dy_2 \\
&= -\frac{2}{\pi}\int_{0}^{\infty} e^{-y_2^2/\sigma^2} dy_2 \\
&\text{Let } v = \frac{y_2^2}{\sigma^2} \implies dv = \frac{2y_2}{\sigma^2} dy_2,\ y_2 = \sigma\sqrt{v}\,(\because y_2 \in (0,\infty)) \\
&= -\frac{2}{\pi}\int_{0}^{\infty} \frac{\sigma^2}{2\sigma\sqrt{v}} e^{-v} dv \\
&= -\frac{\sigma}{\pi}\int_{0}^{\infty} v^{-1/2} e^{-v} dv \\
&= -\frac{\sigma}{\pi}\Gamma\left(\frac{1}{2}\right) \\
&= -\frac{\sigma}{\sqrt{\pi}}
\end{aligned}
$$

(b)

Consider the relationship between the original sample $\{X_1, X_2\}$ and the order statistics $\{Y_1, Y_2\}$. Since $\{Y_1, Y_2\}$ is just a permutation of $\{X_1, X_2\}$, their product remains the same:

$$Y_1Y_2 = X_1X_2$$

Since X_1 and X_2 are independent samples from $N(0, \sigma^2)$, we have:

$$\mathbb{E}[Y_1Y_2] = \mathbb{E}[X_1X_2] = \mathbb{E}[X_1]\mathbb{E}[X_2] = 0 \cdot 0 = 0$$

Additionally, the $N(0, \sigma^2)$ distribution is symmetric about 0, which implies:

$$\mathbb{E}[Y_2] = -\mathbb{E}[Y_1]$$

Therefore, the covariance of Y_1 and Y_2 is:

$$Cov(Y_1, Y_2) = \mathbb{E}[Y_1Y_2] - \mathbb{E}[Y_1]\mathbb{E}[Y_2] = 0 - \left(-\frac{\sigma}{\sqrt{\pi}}\right)\left(\frac{\sigma}{\sqrt{\pi}}\right) = \frac{\sigma^2}{\pi}$$

4.4.19

Because X and Y independent :

$$f_{X,Y}(x, y) = f_X(x) \cdot f_Y(y) = (2x)(3y^2) = 6xy^2, \quad 0 < x, y < 1$$

Then :

$$
\begin{aligned}
f_{U,V}(u, v) &= f_{X,Y}(u, v)|J| + f_{X,Y}(v, u)|J| \\
&= 6uv^2 \cdot 1 + 6vu^2 \cdot 1 \\
&= 6uv(u + v)
\end{aligned}
$$

$$\Rightarrow f_{U,V}(u, v) = \begin{cases} 6uv(u + v), & 0 < u < v < 1 \\ 0, & \text{elsewhere} \end{cases}$$

5.2.2

$$\begin{aligned}
F_{Y_1}(y) &= 1 - P(y < min(X_1, X_2, \dots X_n)) \\
&= 1 - \left(\int_y^\infty e^{-(x-\theta)}\, dx\right)^n \\
&= 1 - \left(\left[-e^{-(x-\theta)}\right]_y^\infty\right)^n \\
&= 1 - \left(e^{-(y-\theta)}\right)^n \\
&= 1 - e^{-n(y-\theta)}, \text{ where } y \in (\theta, \infty)
\end{aligned}$$

Then :

$$\begin{aligned}
F_{Z_n}(z) &= P(Z_n \le z) \\
&= P(n(Y_1 - \theta) \le z) \\
&= P(Y_1 \le \theta + \frac{z}{n}) \\
&= 1 - e^{-z}, \text{ where } z \in (0, \infty)
\end{aligned}$$

$$\lim_{n\to\infty} F_{Z_n}(z) = \lim_{n\to\infty} 1 - e^{-z} = 1 - e^{-z}, \text{ where } z \in (0, \infty)$$

Therefore we have :

$$Z_n \xrightarrow{D} \text{Exponential}(1)$$

5.3.3

$$\mathbb{E}[Y] = 72 \times \frac{1}{3} = 24, \quad Var(Y) = 72 \times \frac{1}{3} \times \frac{2}{3} = 16$$

Then we need continuity correction :

$$\begin{aligned}
P(22 \le Y \le 28) &\approx \Phi(\frac{28.5 - 24}{\sqrt{16}}) - \Phi(\frac{21.5 - 24}{\sqrt{16}}) \\
&\approx \Phi(1.125) - \Phi(-0.625) \\
&\approx 0.8697 - 0.2660 \\
&\approx 0.6037
\end{aligned}$$

6.1.11

(a)

The pdf of $N(\theta, \sigma^2)$:

$$\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\theta)^2/2\sigma^2}$$

The log likelihood function :

$$l(\theta) = \sum_{i=1}^n \left[-\frac{1}{2}\ln(2\pi\sigma^2) - \frac{(x_i - \theta)^2}{2\sigma^2}\right]$$

$$\frac{\partial l(\theta)}{\partial \theta} = \sum_{i=1}^n \frac{x_i - \theta}{\sigma^2} = 0 \implies \theta = \frac{1}{n}\sum_{i=1}^n x_i$$

$$\frac{\partial^2 l(\theta)}{\partial \theta^2} = -\frac{n}{\sigma^2} < 0$$

Therefore we have maximum likelihood function when $\theta = \hat{\theta}$ where :

$$\hat{\theta} = \frac{1}{n}\sum_{i=1}^n x_i = \bar{X}$$

(b)

- **Case 1:** $\bar{X} \geq 0$:
  By, (a) we had proved that when $\hat{\theta} = \bar{X}$ likelihood function has maximum.

- **Case 2:** $\bar{X} < 0$
  The global maximum is at $\bar{X}$, which is outside the feasible region. For $\theta \geq 0$, we have $\theta > \bar{X}$. Since $l(\theta)$ is increasing for $\theta < \bar{X}$ and decreasing for $\theta > \bar{X}$, the function $l(\theta)$ is **strictly decreasing** on the interval $[0, \infty)$. Therefore, the maximum on this interval occurs at the lower boundary:

$$\hat{\theta}_{MLE} = 0$$

Combining both cases, the restricted MLE is:

$$\hat{\theta} = \max\{0, \bar{X}\}$$